Local BH-BH merger rate density [$\mathrm{Gpc^{-3}\ yr^{-1}}$]
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
10^{4}
10^{5}
Gravitational waves
GWTC-2 (Abbott et al. 2020)
GWTC-3 (Abbott et al. 2021)
GWTC-4 (Abbott et al. 2025)
GWTC-5 (Abbott et al. 2026)
Isolated binary evolution
O'Shaughnessy et al. (2010)
Mennekens and Vanbeveren (2014)
Dominik et al. (2015)
de Mink and Belczynski (2015)
Lamberts et al. (2016)
Lipunov et al. (2017)
Mapelli et al. (2017)
Ablimit and Maeda (2018)
Chruslinska et al. (2018)
Giacobbo and Mapelli (2018)
Giacobbo and Mapelli (2020)
Klencki et al. (2018)
Kruckow et al. (2018)
Mapelli and Giacobbo (2018)
Artale et al. (2019)
Baibhav et al. (2019)
Chruslinska et al. (2019)
Eldridge et al. (2019)
Neijssel et al. (2019)
Spera et al. (2019)
Belczynski et al. (2020)
Santoliquido et al. (2020)
Tang et al. (2020)
Zevin et al. (2020)
Bavera et al. (2021)
Bouffanais et al. (2021)
Ghodla et al. (2021)
Mapelli et al. (2021)
Olejak et al. (2021)
Riley et al. (2021)
Roman-Garza et al. (2021)
Santoliquido et al. (2021)
Shao and Li (2021)
Briel et al. (2022b)
Broekgaarden et al. (2021, 2022)
Dorozsmai and Toonen (2022)
Olejak et al. (2022)
Stevenson and Clarke (2022)
van Son et al. (2022)
van Son (2022) SMT only
Hendriks et al. (2023)
Romagnolo et al. (2023)
Srinivasan et al. (2023)
van Son et al. (2023)
Boesky et al. (2024)
Rauf et al. (2024)
de Sá et al. (2024)
Li et al. (2025)
Marinacci et al. (2026)
Romagnolo et al. (2025)
Sgalletta et al. (2025)
Smith and Kaplinghat (2025)
Arca Sedda et al. (2026)
Boco et al. (2026)
Levina et al. (2026)
Sgalletta et al. (2026)
Smith et al. (2026)
Chemically homogeneous evolution
Mandel and de Mink (2016)
Marchant et al. (2016)
Riley et al. (2021)
du Buisson et al. (2020)
Stevenson and Clarke (2022)
Li et al. (2025)
Population III stars
Kinugawa et al. (2014)
Belczynski et al. (2017)
Kinugawa et al. (2020)
Hijikawa et al. (2021)
Liu and Bromm (2021)
Tanikawa et al. (2021)
Triples / Multiples
Antonini et al. (2017)
Silsbee and Tremaine (2017)
Rodriguez and Antonini (2018)
Martinez et al. (2020)
Trani et al. (2021)
Vynatheya and Hamers (2021)
Stegmann et al. (2022)
Wide isolated binaries + flybys
Raveh et al. (2022)
Globular clusters
Bae et al. (2014)
Rodriguez et al. (2015)
Antonini and Rasio (2016)
Rodriguez et al. (2016a)
Askar et al. (2017)
Fujii et al. (2017)
Park et al. (2017)
Fragione and Kocsis (2018)
Hong et al. (2018)
Rodriguez and Loeb (2018)
Choksi et al. (2019)
Kremer et al. (2020)
Antonini and Gieles (2020)
Mapelli et al. (2021)
Nuclear star clusters
Miller and Lauburg (2009)
Antonini and Perets (2012)
Antonini and Rasio (2016)
Bartos et al. (2017)
Petrovich and Antonini et al. (2017)
Stone et al. (2017)
Hamers et al. (2018)
Hoang et al. (2018)
Fragione et al. (2019)
Rasskazov and Kocsis (2019)
Stephan et al. (2019)
Arca Sedda (2020a)
Gröbner et al. (2020)
Tagawa et al. (2020)
Wang et al. (2021)
Yang et al. (2020)
Ford and McKernan (2021)
Mapelli et al. (2021)
Bökholt et al. (2022)
Codazzo et al. (2022)
Young / Open star clusters
Ziosi et al. (2014)
Mapelli et al. (2016)
Rastello et al. (2019)
Di Carlo et al. (2020)
Kumamoto et al. (2020)
Mapelli et al. (2021)
Santoliquido et al. (2020)
Banerjee (2021)
Rastello et al. (2021)
Primordial
Bird et al. (2016)
Ali-Haïmoud et al. (2018)
Raidal et al. (2019)
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
10^{4}
10^{5}
Local BH-BH merger rate density [$\mathrm{Gpc^{-3}\ yr^{-1}}$]